

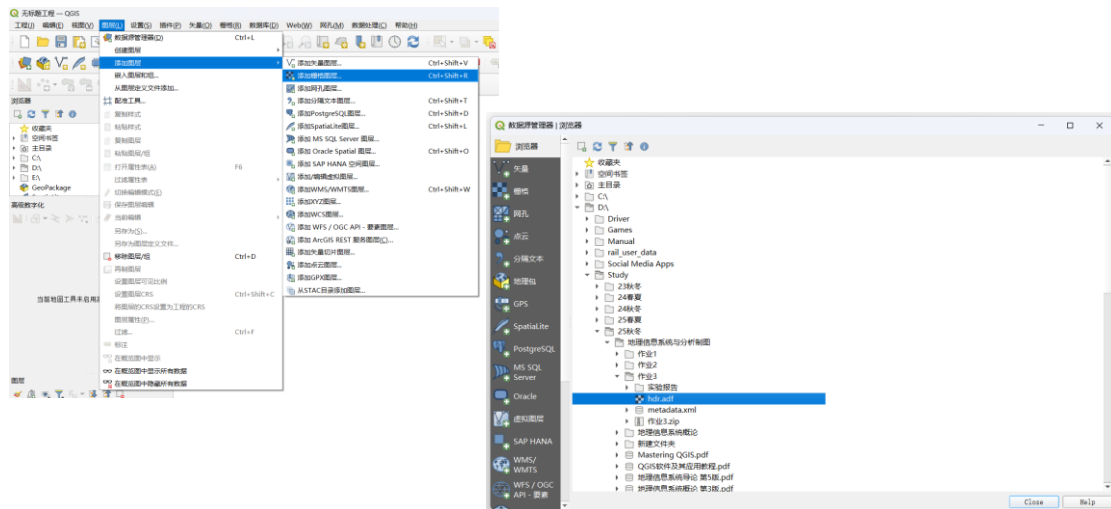
实验三 空间分析 – 数字地形分析

一、实验设备、实验环境

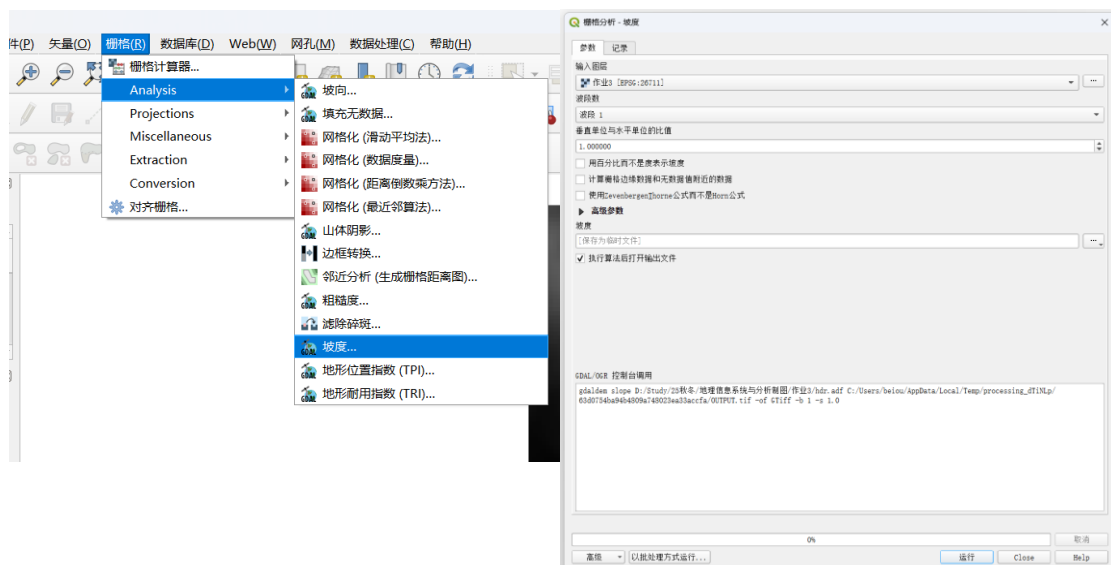
采用 QGIS 3.44 版本的笔记本一台

二、实验步骤

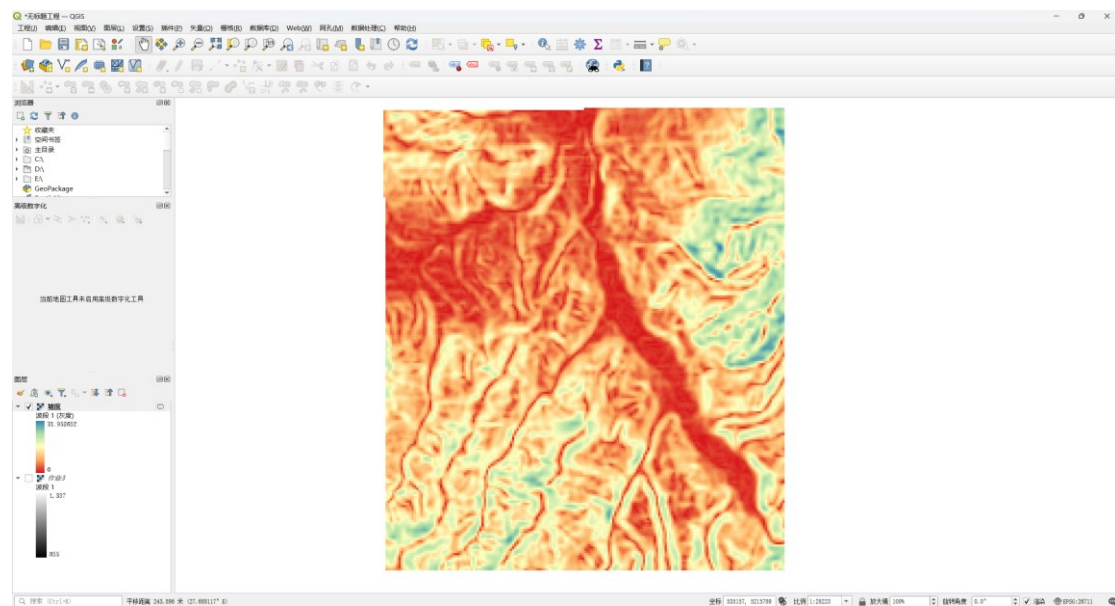
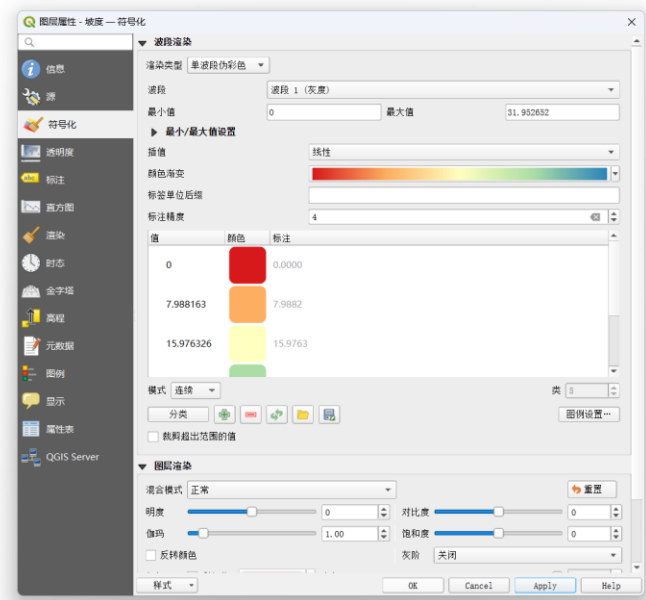
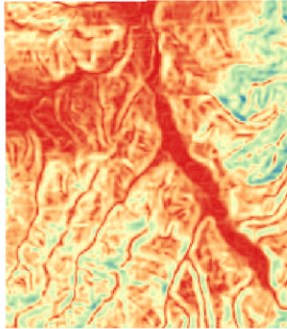
1、打开 QGIS 软件，在图层中选择“图层”-“添加图层”-“添加栅格图层”，在新页面选择浏览器后选中 hdr.adf，点击上方“添加选中的图层”。



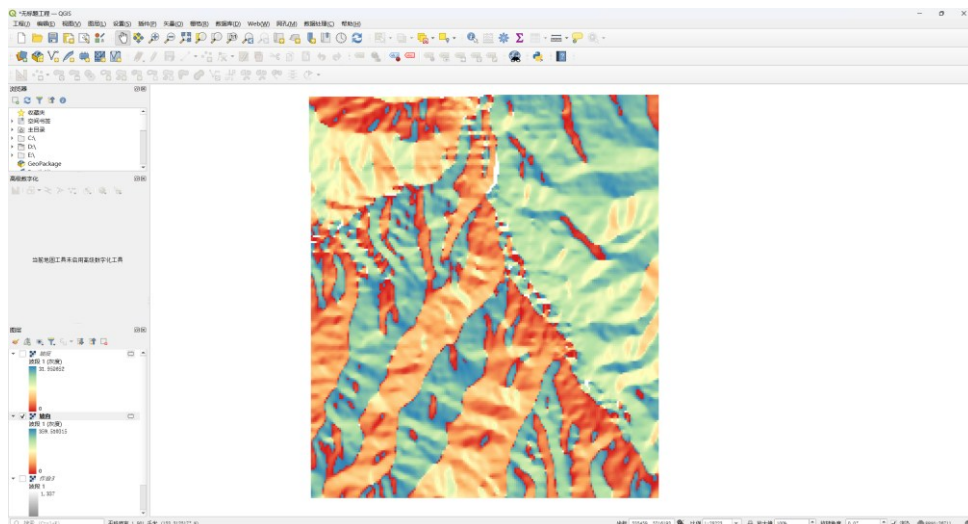
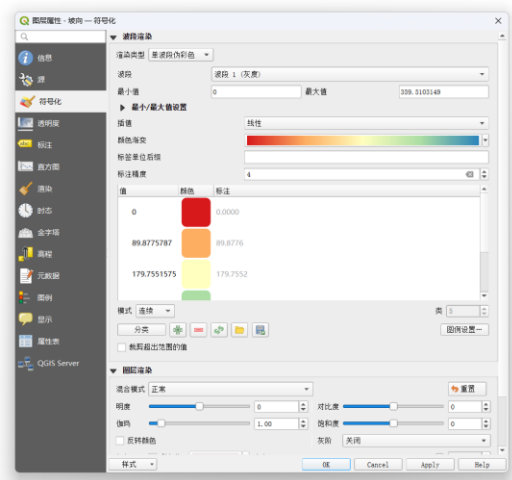
2、菜单栏选中“栅格”-“Analysis”-“坡度”，按图设置后运行。



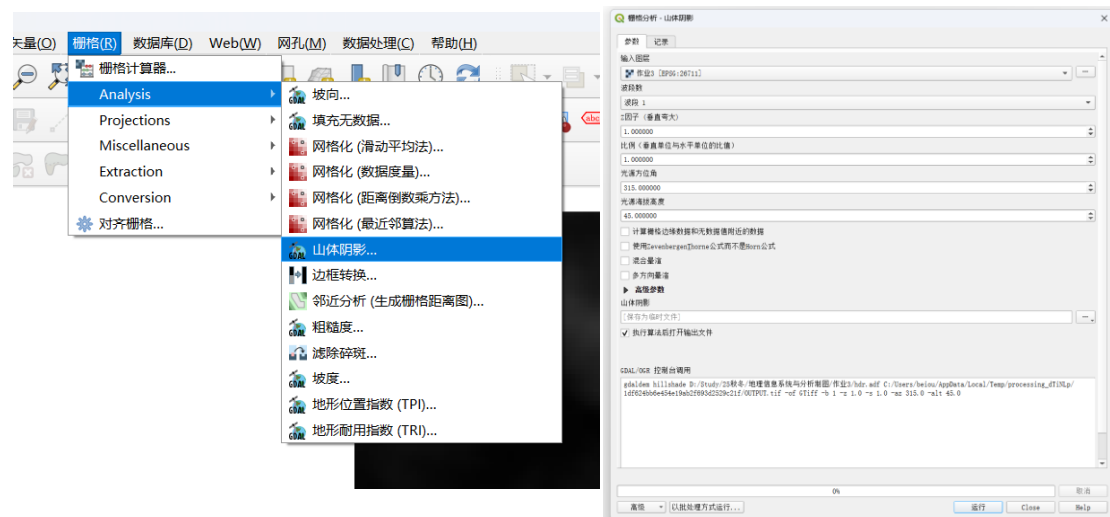
3、运行后会新建一个黑白的坡度图层，右键该图层选择“属性”-“符号化”，按照下图所示设置后即可完成坡度图。



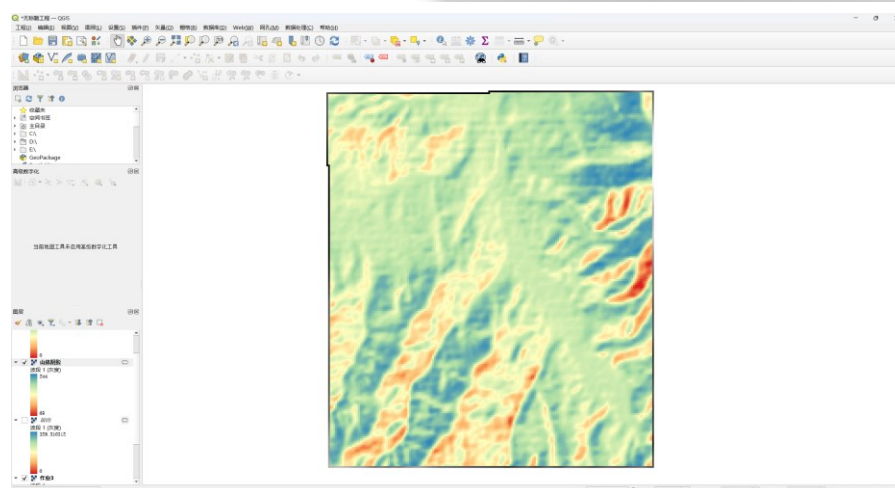
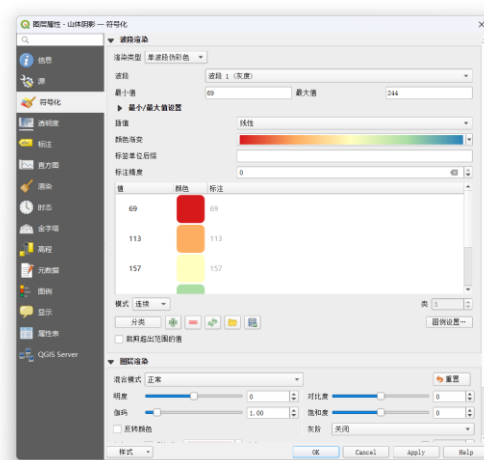
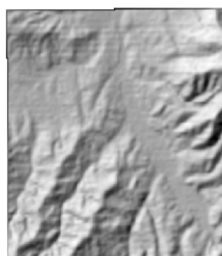
The screenshot shows the QGIS application window. The 'Processing' menu is open, displaying a list of tools including 'Analysis', 'Projections', 'Miscellaneous', 'Extraction', 'Conversion', 'Align Raster...', 'Fill No Data...', 'Grid (Moving Average)...', 'Grid (Data Measure)...', 'Grid (Distance to Nearest Method)...', 'Grid (Nearest Neighbor)...', 'Shaded Relief...', 'Edge Detection...', 'Neighborhood (Generate Distance Map)...', 'Slope...', 'Filter Raster...', 'Slope...', 'Topographic Position Index (TPI)...', and 'Topographic Relief Index (TRI)'. The 'Processing Toolbox' panel on the right is also visible, showing a search bar and a list of processing algorithms.



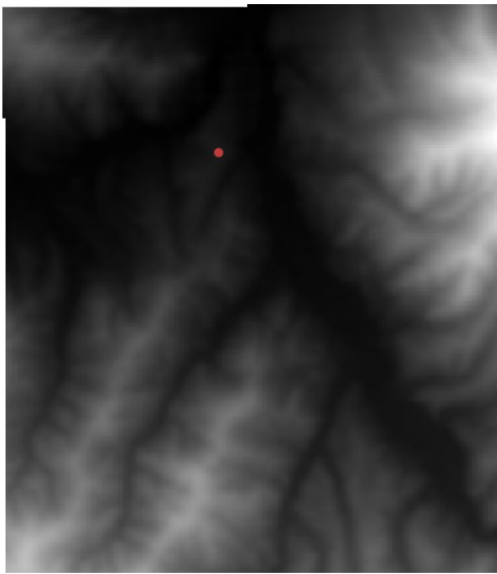
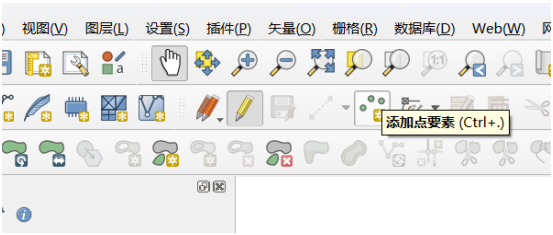
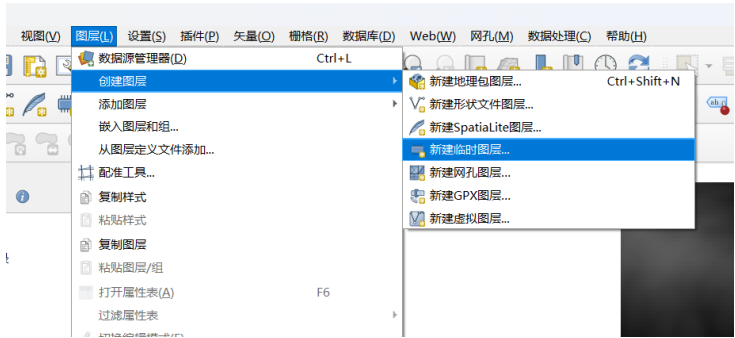
6、菜单栏选中“栅格”-“Analysis”-“山体阴影”，按图设置后运行。方位角默认 315.0，天顶角默认 45.0。



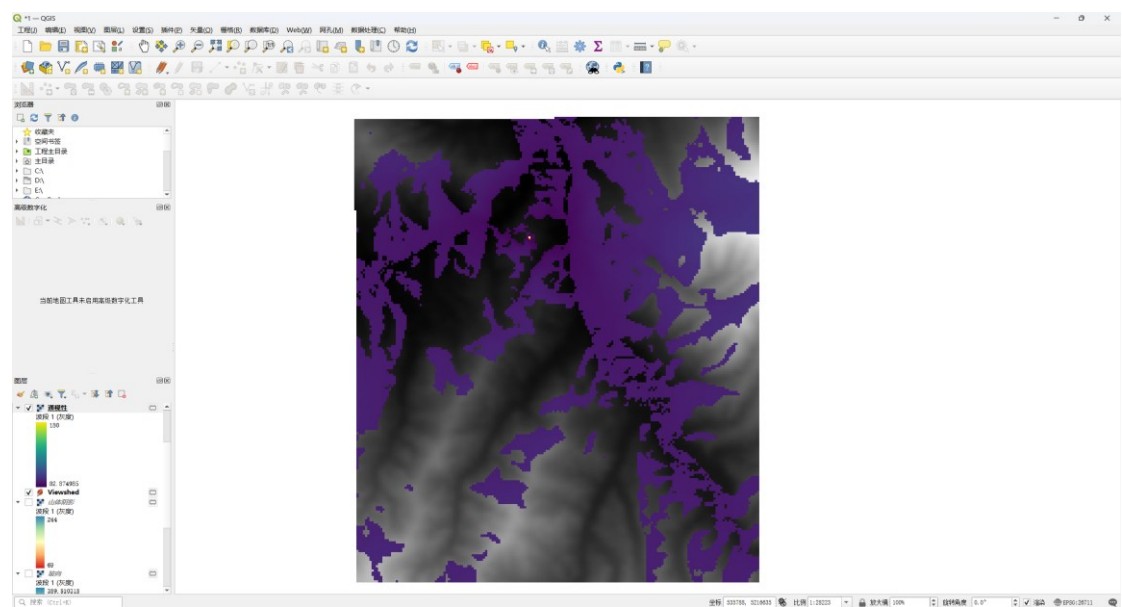
7、运行后会新建一个黑白的山体阴影图层，右键该图层选择“属性”-“符号化”，按照下图所示设置后即可完成坡度图。



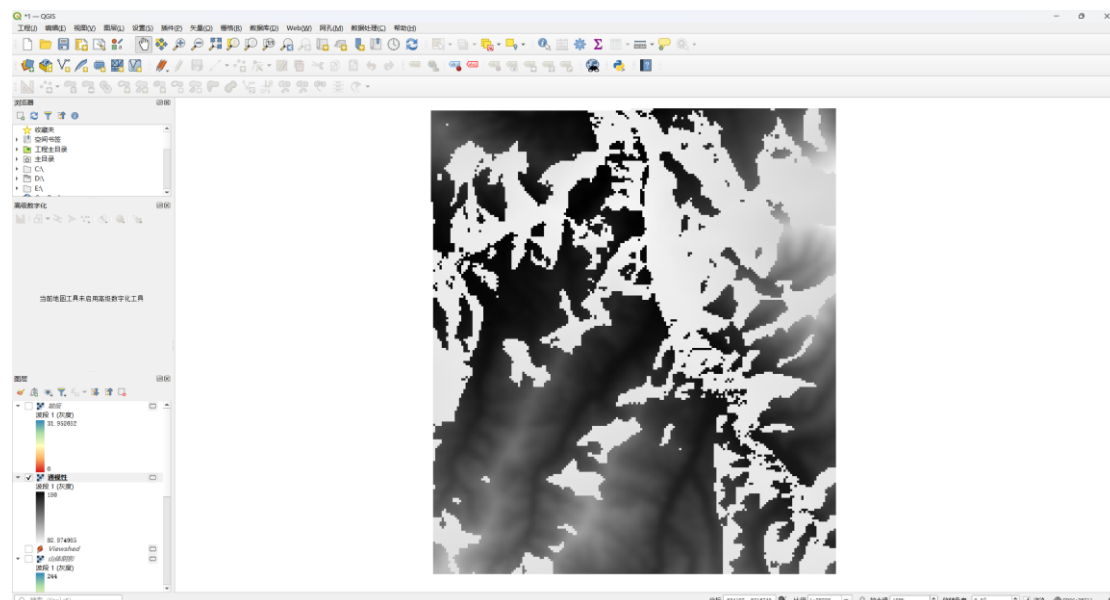
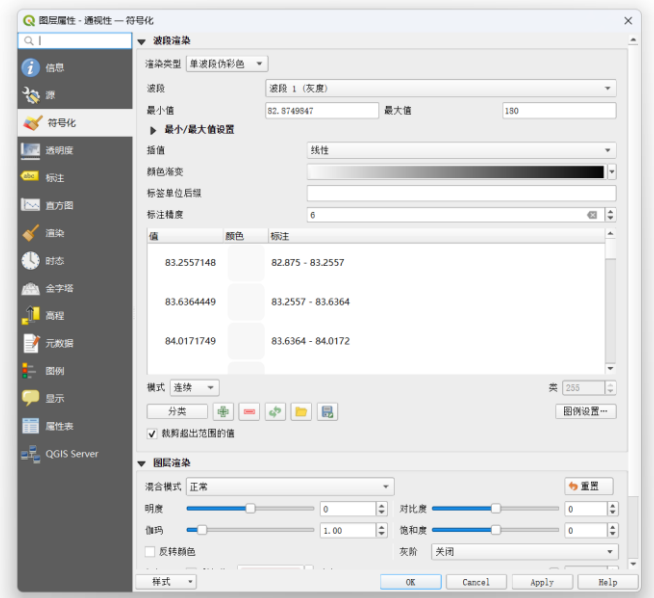
8、在图层中选择“图层” - “创建图层” - “新建临时图层”，按图设置。完成后在 Viewshed 图层打开编辑模式，选择“添加点要素”，选择一个点作为观察点（坐标在第 9 步给出）。



9、在菜单栏选择“数据处理”-“工具栏”，搜索“r.viewshed”，按图设置，点击运行，结果如下。



10、运行后会新建一个通透性图层，右键该图层选择“属性”-“符号化”，按照下图所示设置后即可完成可视区域图。



11、对于每一个做好的图层（坡度、坡向、山体阴影、Viewshed），右键图层名-“导出”-“另存为”，文件名自行设置即可。

The screenshot shows the 'Export Layer as Raster' dialog box in QGIS. The dialog is titled '栅格图层另存为...' (Export Layer as Raster...). It contains several sections for configuring the export process.

Output Mode: The 'Original Data' (原始数据) radio button is selected, and the 'Render Image' (渲染图像) radio button is unselected.

Format: The 'Format' (格式) dropdown menu is set to 'GeoTIFF'. There is a checkbox for 'Create VRT' (创建VRT) which is currently unchecked.

Filename: The 'Filename' (文件名) field is empty, with a browse button ('...') to its right.

Layer Name: The 'Layer Name' (图层名) field is empty.

CRS: The 'CRS' (CRS) dropdown menu is set to 'EPSG:26711 - NAD27 / UTM zone 11N'.

Range (Current: Layer): This section allows specifying the spatial extent of the raster. The coordinates are: North (北) 5218935.0000, West (西) 528435.0000, East (东) 534015.0000, and South (南) 5212515.0000. Below these are three dropdown menus for 'Calculate from' (计算从): 'Layer' (图层), 'Layout Map' (布局地图), and 'Bookmark' (书签). At the bottom of this section are two buttons: 'Current Layer Range' (当前图层范围) and 'Map Canvas Range' (地图画布范围).

Resolution (Current: Layer): This section allows specifying the resolution of the raster. The 'Horizontal' (水平) radio button is selected, with a value of 30. The 'Vertical' (垂直) radio button is unselected, with a value of 30. There are buttons for 'Layer Resolution' (图层分辨率) and 'Layer Size' (图层大小). Below these are two more input fields: 'Columns' (列) with a value of 186 and 'Rows' (行) with a value of 214.

Create Options: This section is currently collapsed. It contains a 'Configuration' (配置) dropdown menu set to 'Default' (默认). Below this is a table with two columns: 'Name' (名称) and 'Value' (值).

Footer: At the bottom of the dialog, there is a checkbox 'Add saved files to map' (将已保存的文件添加到地图) which is checked. To the right of this checkbox are three buttons: 'OK', 'Cancel', and 'Help'.